

Mohammad Amin Sadat

Mechatronics Engineer/AI Researcher



Iran



s.mohammad.amin.sadat@gmail.com



MohammadAminSadat



in LinkedIn

Education

- MS** **Shahid Beheshti University**, Mechatronics Engineering Sep 2022 – Feb 2025
- GPA: 17.54/20 (Received an Excellent Score for Master's Thesis)
 - Highest GPA in Mechatronics Engineering
 - Thesis: "Path tracking control of a self-driving car using model predictive control and reinforcement learning"
- BS** **Shiraz University**, Mechanical Engineering Sep 2017 – Feb 2022
- GPA: 14.68/20

Skills

Python
MATLAB
SolidWorks
LaTeX
Linux
PyTorch



Embedded Systems
ROS
Altium Designer
C/C++
Rust
TensorFlow



Test Scores

IELTS Overall: 7.0 Speaking: 6.5 Listening: 7.5 Reading: 7.0 Writing: 6.0

GRE Scheduled for May 2025

Research Interests

- Control Theory Research
- Model Predictive Control
- AI/ML/DL/Reinforcement Learning
- Fuzzy Logic and Control
- Learning Based Control
- Mechatronics, Robotics, and Autonomous Vehicles
- Navigation in Autonomous Systems

Publications

- Path tracking control of a self-driving car using model predictive control and reinforcement learning - Master's Thesis** Feb 2025
- Deep Reinforcement Learning-Assisted Nonlinear Model Predictive Control for Improved Path Tracking in Autonomous Vehicles** Ready To Submit
- S.M.Amin Sadat, Hadi Asharioun, Mostafa Taghizadeh, Mahmood Mazare*

Experience

- Teaching Assistant** Shahid Beheshti University & Shiraz University
- Courses:** Statics, Automatic Control, Artificial Intelligence, Modern Control
 - Responsibilities:** Taught lectures, designed projects (MATLAB/Python), and graded assignments
- Avid Mechanic Hafez Sanaat**, Mechatronics Engineer AI Engineering Oct 2021 - Mar 2023
- Developed software, Mechanical Design and PCB design for a hospital bed and a robotics manipulator arm
 - Participating in AI projects
- Software Motion**, ADAS Engineer at DPC (Decision, Planing & Control) team Apr 2025 - Present
- Developing ADAS systems for vehicles

Volunteering

Core Member of Robotics Club

- Instructor for ARDUINO and MATLAB programming courses

Shiraz University
Sep 2019 - Sep 2020

Certifications

- Generative AI for everyone (Coursera) [🔗](#)
- Motion Planing for Self-Driving Cars (Coursera) [🔗](#)
- State Estimation and Localization for Self-Driving cars (Coursera) [🔗](#)

Honor and Awards

- Full tuition waiver scholarship for Bachelor's and Master's degree.
- Top 2% in the national BSc entrance exam in the felid of Mathematics and Physics in 2017 in Iranian Universities.
- Top 0. 5% in the national MSc entrance exam in the felid of mechanical Engineering in 2023 in Iranian Universities.
- Rank 1 in GPA among students in mechatronic engineering at Shahid Beheshti University.

Relevant Coursework

- Modern Control
- Artificial Intelligence
- Introduction to Mechatronics
- Mechatronics I&II
- Automatic Control
- Mechanism Design
- Introduction to machine learning
- Robotics
- Computer Aided Design

Projects

Fuzzy Logic Library fuzzy-logic_rs 🔗 • Developed a Fuzzy Logic and Control Library • Implemented using Rust Programming Language • Fully functional for Type I systems • Still in Development	Hydraulics and Pneumatics Spring 2023 • Designed sequence control for a pneumatic system • Simulated in TIA Portal for Siemens PLC
Artificial Intelligence Fall 2023 • Simulated a PINN paper • Programmed in Python with TensorFlow	Bachelor's Degree Final Project Fall 2021 • Designed and built an IoT garden • Developed a web interface and PCB design
Mechatronics II Spring 2023 • Built an inverted pendulum on a cart • Controlled using PID and Full-State Feedback	Mechanism Design Spring 2021 • Designed and simulated a wire-wrapping tool • Implemented using SolidWorks and Arduino
Introduction to Machine Learning Spring 2022 • Find the best accuracy for yeast and wine dataset • Program in python and with Scikit-learn Library • Got Highest accuracy in my class	Introduction to Mechatronics Spring 2021 • Designing and build a color detection sensor • PCB design • program in Arduino
	Robotics Project Fall 2020 • Designed a convertible quadrotor • Modeled in SolidWorks

References

Available upon request.